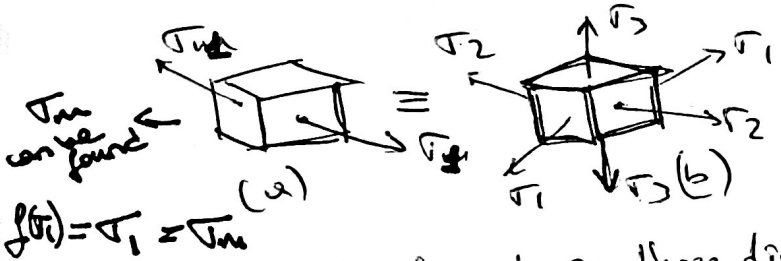
YIELD AND FAILURE CRITERIA (1 week)

Material tests provides a stress value which is used to determine if the materials are in their elastic limits or failed.

However, conducting three dimensional material testson the materials are not suitable since there are infinite number of test configurations

Hence, only one-dimensional test can be directly used to determine failure limits. So, we need a relation between uniaxial and triaxial stress states.

Failure criteria relate the one (uniaxial) dimensional stress state to the three (triaxial) dimensional stress-state considering failure limits



$\rightarrow \sigma_{VM} = f(\sigma_1, \sigma_2, \sigma_3)$

(a) and (b) both attain safety limits

If the material in three dimensional stress-state does not yield then;

$\sigma_{VM} > f(\sigma_1, \sigma_2, \sigma_3)$

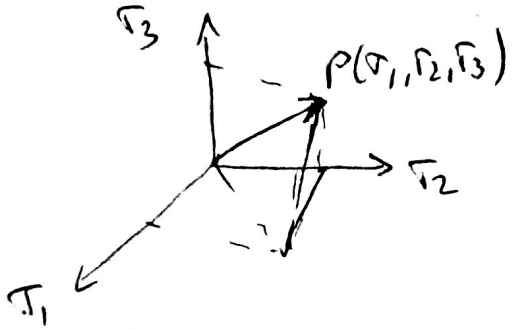
* σ_{VM} can be determined from uniaxial tension and compression tests

Different $f(\sigma_1, \sigma_2, \sigma_3)$ functions were proposed based on which quantity is used to define the failure (safety) limit.

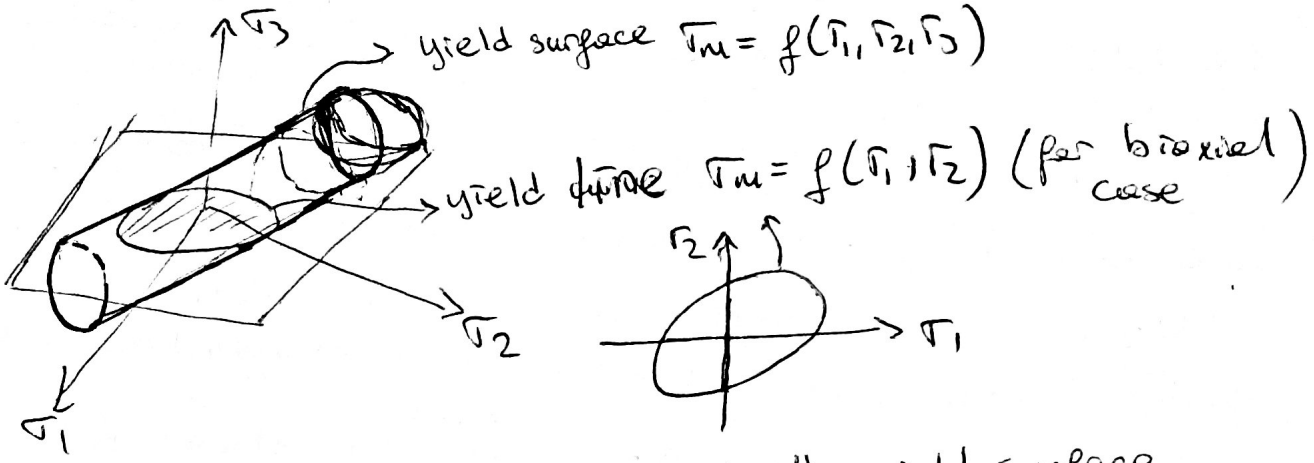
- Stress-based
- Energy-based

(Strain based criteria are not used anymore)

Principle stress-space: Any stress state which is given by three principal stresses of it, defines a point in this space.



For every stress values $\sigma_1, \sigma_2, \sigma_3$ of which the material attains the safety limit, one can draw a yield surface.



if the considered stress-state is inside the yield surface then it can be said that the material is <u>not</u> in dangerous state. and vice versa.

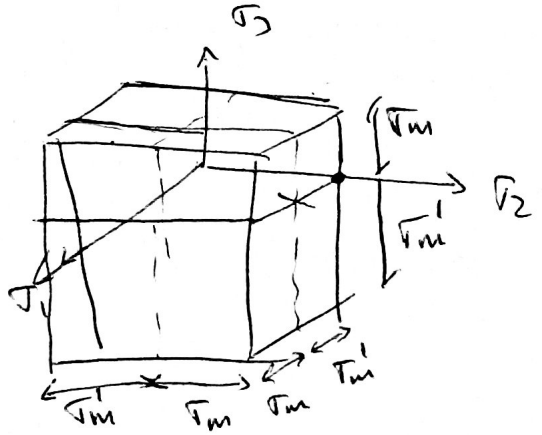
<u>Stress-Space</u> : In this case Mohr's circle convention is applied.

Stress-Based Criteria

These criteria hold the stresses responsible for failure

<u>The maximum normal stress criterion :</u>

It is assumed that the max. principal stress causes material failure or yield.

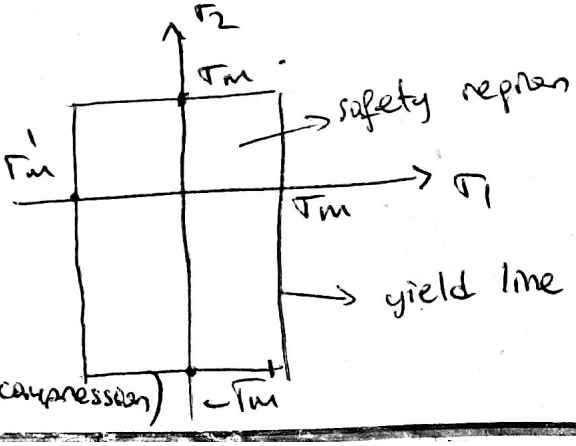


$\sigma_{max} = \sigma_1 = \sigma_M$ or $\sigma_{min} = |\sigma_3| = \sigma_M'$

↓ yield stress at uniaxial tension

↓ yield stress at uniaxial compression

In two dimensional stress state



* Suitable for brittle materials
* does not satisfy hydrostatic compression test (Materials cannot experience failure under hydrostatic compression)

Maximum Shear Stress Criterion: (Tresca)

It is assumed that the max. shear stress in triaxial case causes material failure or yielding

$$\tau_{max} = \frac{1}{2}(\sigma_1 - \sigma_3)$$

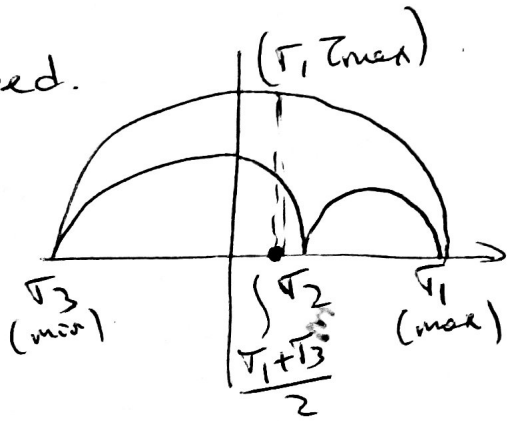
$\sigma_1 \rightarrow$ max. principle stress
$\sigma_3 \rightarrow$ min. " "



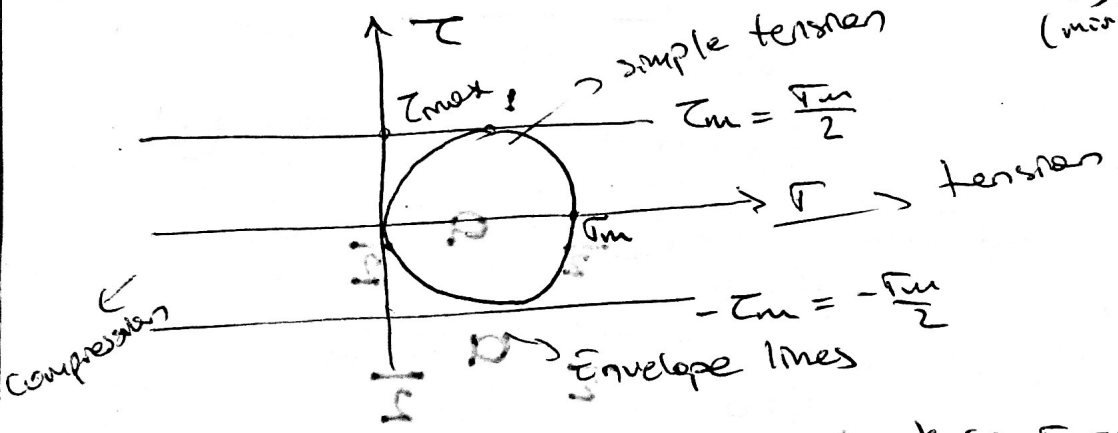
In uniaxial case $\tau_m = \frac{\sigma_m}{2}$, Hence

$$\tau_{max} = \tau_m \rightarrow \boxed{\sigma_1 - \sigma_3 = \sigma_m}$$

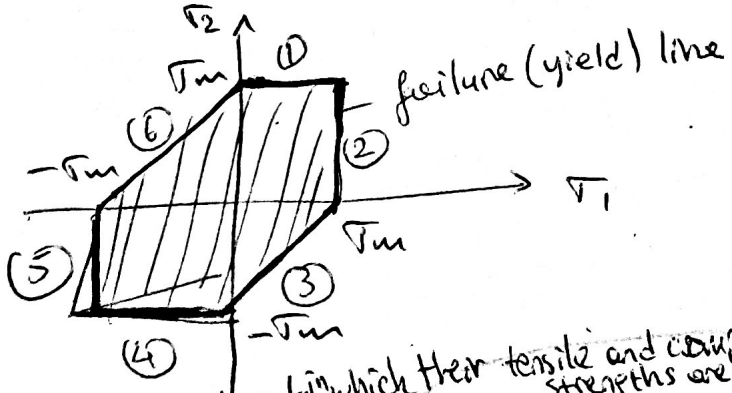
- The intermediate stress is not considered. (σ_2)

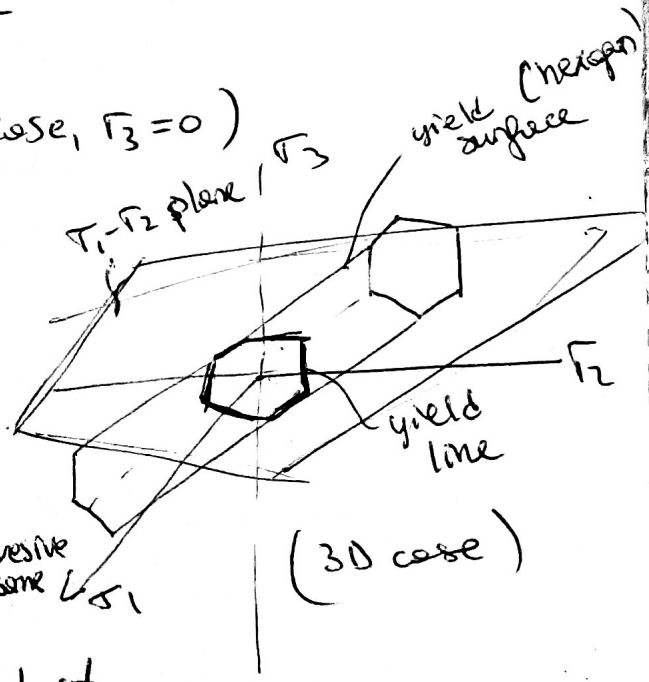


In Stress-space



In principle stress-space (2D case, $\sigma_3 = 0$)





(3D case)

* Suitable for materials in which their tensile and compressive strengths are same
* Suitable for ductile materials
* Satisfy the hydrostatic compression test
* does not satisfy the hydrostatic tension test

Coulomb Shear Strain Criterion

Although its assumed that shear strain is responsible for failure, this criterion takes the effect of normal stress on the plane in which the shear stress τ_{max} acts.



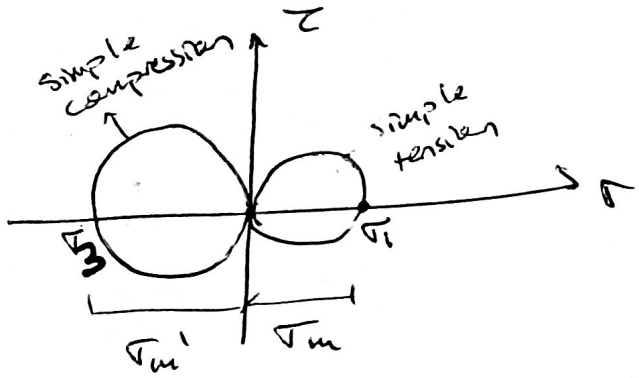
Benefitting from internal friction law, it can be written. shear strength is linear function of σ

$$\tau_{max} = a - b\sigma \quad (1)$$

σ is the normal stress on the plane in which τ_{max} acts

In principal space

Consider the simple compression and tension cases. in stress-space



Solve for a and b in principal stress-space

$$\tau_{max} = \frac{1}{2}(\sigma_1 - \sigma_3) \quad (2)$$

$$\sigma = \frac{1}{2}(\sigma_1 + \sigma_3) \quad (3)$$

$$\tau_{max} = \tau_m = a - b\sigma_m$$

Substituting (2),(3) into (1)

$$\frac{1}{2}(\sigma_1 - \sigma_3) = a - \frac{b}{2}(\sigma_1 + \sigma_3) \quad (4)$$

$\tau_m = a - b\sigma \Longrightarrow$

Simple Tension: $\sigma_1 = \sigma_m$, $\sigma_3 = 0$, $\tau_{max} = \frac{\sigma_m}{2} = a - \frac{b}{2}\sigma_m$

Simple Comp: $\sigma_3 = \sigma_m'$, $\sigma_1 = 0$, $\tau_{max} = \frac{\sigma_m'}{2} = a + \frac{b}{2}\sigma_m'$

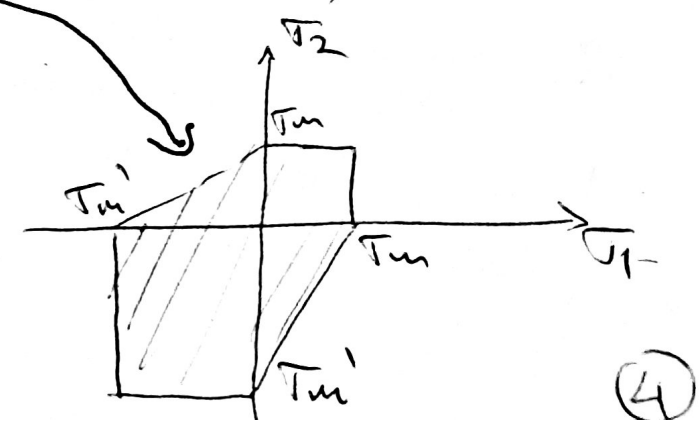
Solve for a and b

$$a = \frac{\sigma_m' \sigma_m}{\sigma_m' + \sigma_m} \qquad b = \frac{\sigma_m' - \sigma_m}{\sigma_m' + \sigma_m}$$

Substitute a and b into 4 and rearrange

$$\boxed{\frac{\sigma_1}{\sigma_m} - \frac{\sigma_3}{\sigma_m'} = 1} \qquad (\sigma_1 > 0 \quad \sigma_3 < 0)$$

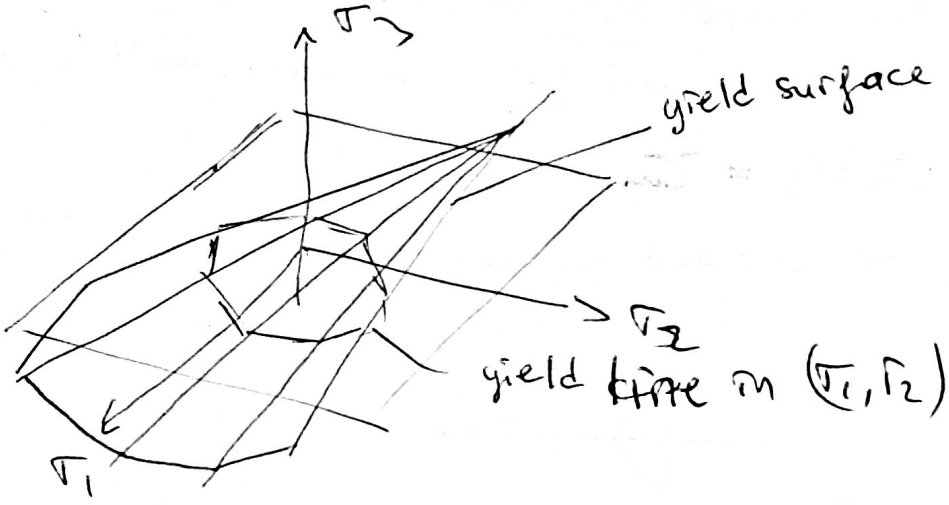
In plane case for $\sigma_m' > \sigma_m$



As a special case if $\sigma_m = \sigma_m'$ the equation will be

$\sigma_1 - \sigma_3 = \sigma_m$ (Tresca criterion)

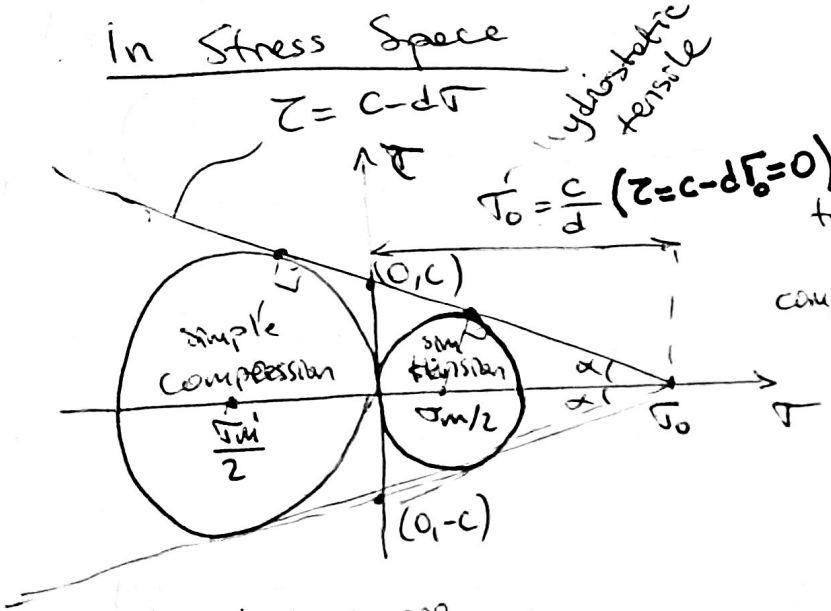
In three dimensional case (principal stress space)



yield trace in (τ_1, τ_2)

In Stress Space

$\tau = c - d\sigma$



$\tau = c - d\sigma$ (1)

Equations of mohr circle

tension $\left(\sigma - \frac{1}{2}\tau_m\right)^2 + \tau^2 = \left(\frac{1}{2}\tau_m\right)^2$

compression $\left(\sigma + \frac{1}{2}\tau_m'\right)^2 + \tau^2 = \left(\frac{1}{2}\tau_m'\right)^2$

rearranging

$\sigma^2 - \tau_m\sigma + \tau^2 = 0$ (2)

$\sigma^2 + \tau_m'\sigma + \tau^2 = 0$ (3)

* If the stresses are inside the envelope lines it is said that the material does not fail

Substitute 1 into 2 and 3

$$\sigma^2 - \left(\frac{\tau_m + 2cd}{1+d^2}\right)\sigma + \frac{c^2}{1+d^2} = 0 \quad (4)$$

$$\sigma^2 + \left(\frac{\tau_m - 2cd}{1+d^2}\right)\sigma + \frac{c^2}{1+d^2} = 0 \quad (5)$$

Since τ line intersects the circles at one point the roots of (4) and (5) must be simple. Hence the discriminant of (4) and (5) must be zero

Using these conditions c and d are found as follows

$$c = \pm\frac{1}{2}\sqrt{\tau_m'\tau_m} \qquad d = \frac{\tau_m' - \tau_m}{2\sqrt{\tau_m'\tau_m}}$$

(discriminant $\Delta = b^2 - 4ac$)

Relation between a, b and c d $\Rightarrow$ $a = c\cos\alpha$ $\tan\alpha = d$

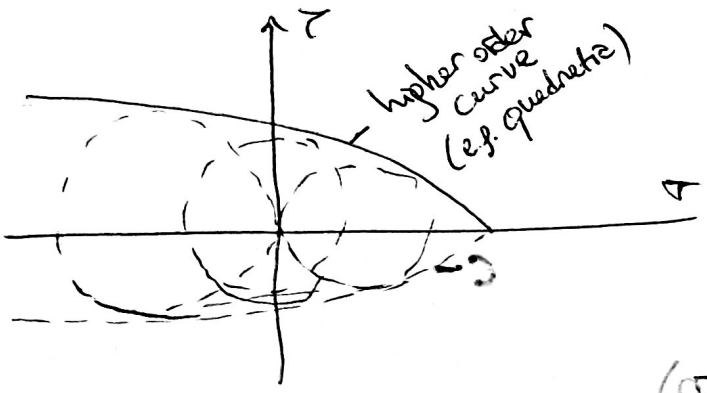
in principal stress space $b = d\cos\alpha$

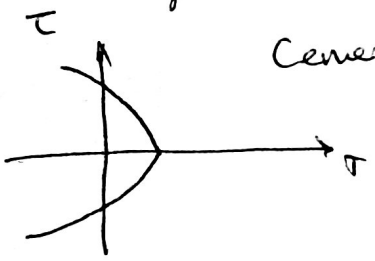
General Mohr shear stress criterion

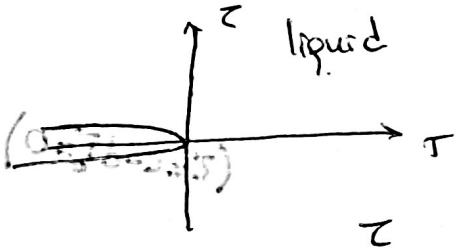
Experiments shows that the envelopes of Mohr circle is not always two straight lines. Hence in general form, it can be written.

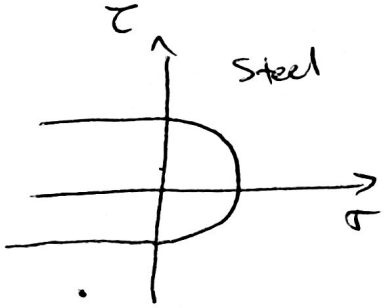
$$\frac{\sigma_1 - \sigma_3}{2} = f(\sigma_1 + \sigma_3) = \tau_m$$

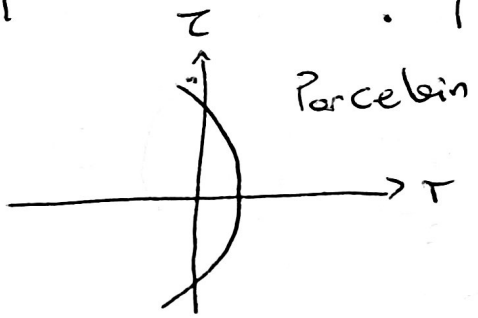
Here the function f is determined for each material



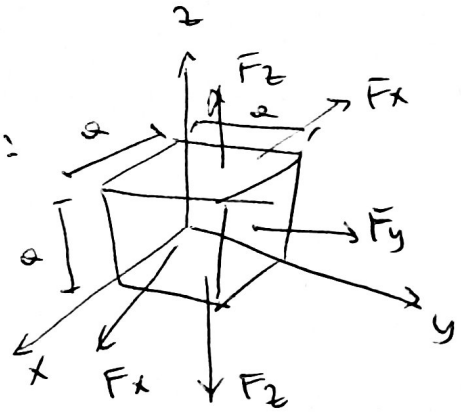








Ex:



$F_x = 50\ kN$ $F_y = 20\ kN$ $F_z = -30\ kN$

σ_m and $\tau_m' = 180\ MPa$

Safety factor $n = 2$ Poisson ratio $\nu = 0.3$

Find the dimension a of a cube considering the safety by

a) Maximum normal stress criterion

b) " shear stress (Tresca) criterion

$$\sigma_{max} = \sigma_1 = \frac{50000}{a^2}$$

$$\sigma_2 = \frac{20000}{a^2}$$

$$\sigma_{min} = \sigma_3 = -\frac{30000}{a^2}$$

$$\sigma_{em} = \frac{\sigma_m}{n} = \frac{180}{2} = 90\ MPa$$

a) Max. normal stress

$\sigma_{max} < \sigma_{em} = 80$ MPa

$$\frac{50000}{a^2} \leq 80 \Rightarrow a \geq 23.6 \text{ mm}$$

b) $\sigma_{max} - \sigma_{min} < \sigma_{em}$

$$\frac{50000}{a^2} - \frac{(-30000)}{a^2} \leq 80 \Rightarrow a \geq 28.8 \text{ mm}$$

Ex: The stresses σ_m and σ_m' obtained from uniaxial tension and compression tests are 20 and 80 MPa, respectively. The principle stresses occur on the material are $\sigma_1 = 12$ MPa, $\sigma_2 = 0$ and $\sigma_3 = -40$ MPa. Determine if the material fails using Coulomb shear strain criterion.

1. way

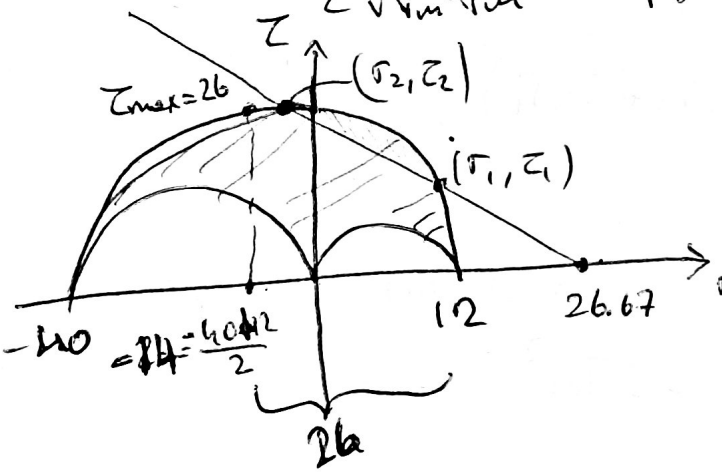
$\tau_n = a - b\sigma$

stress space $\Rightarrow$

$$a = \frac{1}{2}\sqrt{\sigma_m'\sigma_m} = \frac{1}{2}\sqrt{80 \cdot 20} = 20$$

$$b = \frac{\sigma_m' - \sigma_m}{2\sqrt{\sigma_m'\sigma_m}} = \frac{1}{2}\frac{80-20}{\sqrt{80 \times 20}} = \frac{3}{4}$$

$\Rightarrow \tau_n = 20 - \frac{3}{4}\sigma$



for $\tau = 0$ $\sigma = 26.67$

Equation of mohr circle

$$(\sigma - (-14))^2 + \tau^2 = \tau_{max}^2$$

$$(\sigma + 14)^2 + \tau^2 = 26^2$$

$\sigma = \frac{4}{3}(20 - \tau)$ is subtituted into equation

$$25\tau^2 - 976\tau + 8100 = 0 \Rightarrow \begin{cases} \tau_1 = 14.13 \\ \tau_2 = 24.91 \end{cases}$$

corresponding stresses $\sigma_1 = 2.82$

$\sigma_2 = -6.54$

There are two real roots, Hence intersect the circle at two points, material fails!!!

2nd way

$$\frac{\sigma_1}{\sigma_m} - \frac{\sigma_3}{\sigma_m} = 1 \qquad \frac{12}{20} - \frac{(-40)}{80} = \frac{11}{10} \Rightarrow 1.1 > 1$$

fails !!!

Energy Based Criteria

Total Strain Energy criteria

It is assumed that the material fails when its strain energy reaches a certain ~~portion~~ value.

$$u_i = \frac{1}{2E}\left[\sigma_1^2+\sigma_2^2+\sigma_3^2 - 2\nu(\sigma_1\sigma_2+\sigma_2\sigma_3+\sigma_3\sigma_1)\right] = \underbrace{\frac{1}{2E}\sigma_m^2}_{\text{uniaxial case}}$$

u_i = energy stored in unit volume

By simplifying

$$\sigma_1^2+\sigma_2^2+\sigma_3^2 - 2\nu(\sigma_1\sigma_2+\sigma_2\sigma_3+\sigma_3\sigma_1) = \sigma_m^2$$

Does not satisfy the hydrostatic experiment test

Distortion or Shear energy criteria (VON MISES)

Satisfy hydrostatic compression

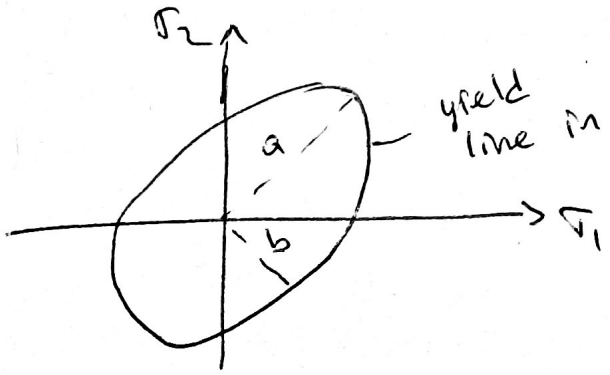
Suitable for ductile materials

$$u_s = \frac{1}{6G}\left(\sigma_1^2+\sigma_2^2+\sigma_3^2 - (\sigma_1\sigma_2+\sigma_2\sigma_3+\sigma_3\sigma_1)\right] = \frac{1}{6G}\sigma_m^2$$

G shear modulus

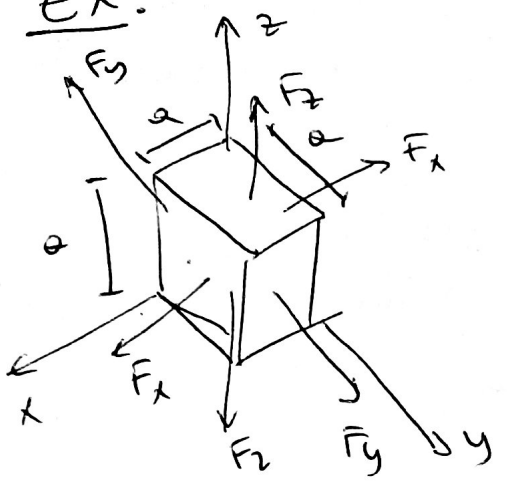
rearranging the expression one can obtain

$$(\sigma_1-\sigma_2)^2 + (\sigma_2-\sigma_3)^2 + (\sigma_3-\sigma_1)^2 = 2\sigma_m^2$$



yield line in $\sigma_1\sigma_2$ plane (ellipse)

EX:



$F_x = 50\,kN$ $F_y = 20\,kN$ $F_z = -30\,kN$

$\sigma_M = \sigma_M' = 180\,MPa$ safety factor $n = 2$

Poisson ratio $\nu = 0.3$

Determine a;

a) Total strain energy criteria

b) Distortion Energy (VON MISES) criteria

$$\sigma_1 = \sigma_{max} = \frac{50000}{a^2} \qquad \sigma_2 = \frac{20000}{a^2} \qquad \sigma_{min} = \sigma_3 = -\frac{30000}{a^2}$$

$$\sigma_{em} = \frac{\sigma_M}{n} = \frac{180}{2} = 90\,MPa$$

a) $$\sigma_1^2 + \sigma_2^2 + \sigma_3^2 - 2\nu(\sigma_1\sigma_2 + \sigma_1\sigma_3 + \sigma_3\sigma_2) \le \sigma_{em}^2$$

$$\frac{10^8}{a^4}\left[5^2 + 2^2 + (-3)^2 - 2 \cdot 0{,}3\,(5 \cdot 2 + 5 \cdot (-3) + (-3) \cdot 2)\right] \le 90^2$$

$$a \ge 27.2\,mm$$

b) $$\sigma_1^2 + \sigma_2^2 + \sigma_3^2 - (\sigma_1\sigma_2 + \sigma_1\sigma_3 + \sigma_2\sigma_3) \le \sigma_{em}^2$$

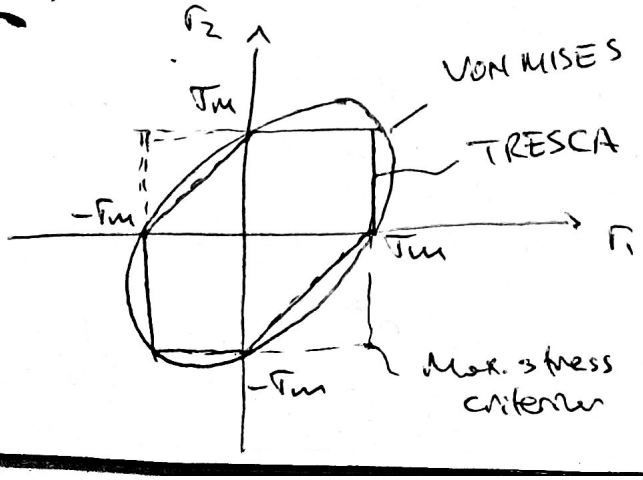
$$\frac{10^8}{a^4}\left(5^2 + 2^2 + (-3)^2 - (5 \cdot 2 + 5(-3) + 3(-3)\right] \le 90^2$$

$$a \ge 27.9\,mm$$

Comparison of Different Criteria

To compare different criteria, it is necessary to overlap failure lines in terms of principal stresses.

for $\sigma_1 - \sigma_2$ plane ($\sigma_3 = 0$)



for ~~Brittle~~ material

→ Caloumb ~~shear~~ stress

→ Mohr general shear stress

for Ductile Material

→ TRESCA

→ VON MISES

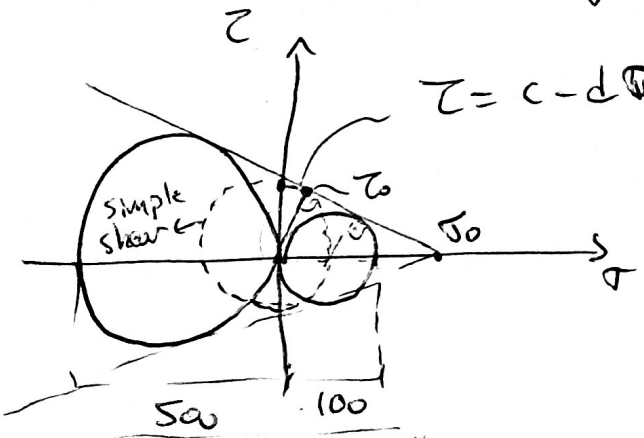
If τ_m and τ_m' are simple tensile and compression strength for $\nu = 0,3$ comparison of different criteria is given as follows

Tresca Criterion	$\tau_m = 0,5\,\tau_m$
Von Mises Criterion	$\tau_m = 0,577\,\tau_m$
Total strain Energy	$\tau_m = 0,62\,\tau_m$
Max. normal stress	$\tau_m = \tau_m$

Ex: Safety stress for simple tensile test is $\tau_m = 100 \frac{N}{cm^2}$ and for compressive test is $\tau_m' = 500 \frac{N}{cm^2}$ for a material

a) Find hydrostatic tensile strength, the equation of envelope lines for Mohr-Coulomb shear stress theory

b) Find the strength of material for simple shear case



By triangle similarity

$$\frac{\tau_0}{50} = \frac{125}{75} \rightarrow (125-50)$$

$$\tau_0 = 83.33 \frac{N}{cm^2}$$

$$c = \frac{1}{2}\sqrt{100\,500} = 111.8034$$

$$d = \frac{500-100}{2\sqrt{500 \cdot 100}} = 0,8944$$

$\tau = c - d\sigma$

$\tau = 111.8034 + 0,8944\,\sigma$

Hydrostatic tensile: for $\tau = 0$

$$\sigma_0 = \frac{c}{d} = \frac{111.803}{0,8944} \approx 125 \frac{N}{cm^2}$$

b) The Mohr's circle equation for simple shear

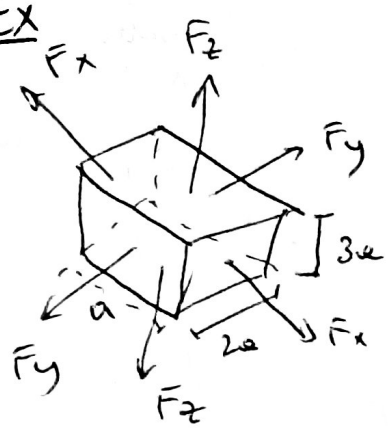
$\sigma^2 + \tau^2 = \tau_0^2$ τ_0 is the simple shear strength

Since the envelope line must be tangent to the circle

$\sigma^2 + (\underbrace{111,8034}_{c} + \underbrace{0,8944}_{d}\sigma)^2 = \tau_0$ → The determinant should be zero

$$\Delta = 4d^2c^2 - 4(1+d^2)(c^2 - \tau_0^2) = 0 \qquad \tau_0 = \frac{c}{\sqrt{1+d^2}} = 83,33 \frac{N}{cm^2}$$

Ex



$F_x = 24$, $F_y = 9$, $F_z = -8$ kN

Considering the fact that the envelope curve of the material is $\tau = 320 - \sigma^2$ [MPa] determine a.

$$\sigma_1 = \frac{24000}{6a^2} = \frac{4000}{a^2} \qquad \sigma_2 = \frac{9000}{3a^2} = \frac{3000}{a^2} \qquad \sigma_3 = -\frac{9000}{a^2} = \frac{-4000}{a^2}$$

$$\tau_{max} = \frac{1}{2}(\sigma_1 - \sigma_3) = \frac{1}{2}\left(\frac{4000}{a^2} - \frac{(-4000)}{a^2}\right) = \frac{4000}{a^2}$$

$\frac{1}{2}(\sigma_1 + \sigma_2) = 0$ (center of Mohr's circle

Equation of Mohr's circle

$$\sigma^2 + \tau^2 = \left(\frac{4000}{a^2}\right)^2$$

Equation for envelope curve

$$\tau = 320 - \sigma^2 \rightarrow \sigma^2 = 320 - \tau$$

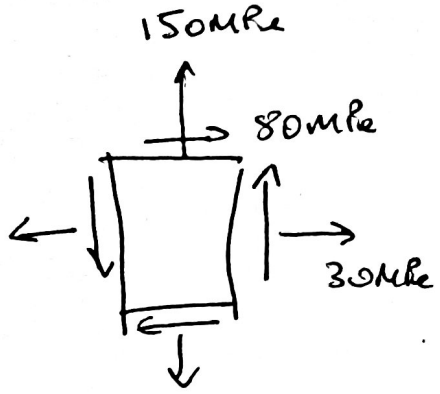
$$\sigma^2 + (320 - \tau) = \left(\frac{4000}{a^2}\right)^2 \Rightarrow \sigma^2 - \tau + 320 - \frac{16\,10^6}{a^4} = 0$$

The root must be single since the curve must be tangent to circle

$$\Delta = b^2 - 4ac$$

$$\Delta = (-1)^2 - 4.1.\left(320 - \frac{16\,10^6}{a^4}\right) = 0 \qquad a \cong \pm 14.96 \text{ mm}$$

The stress-state which belong to a steel element is given in Fig. If $\sigma_m = 350$ MPa calculate the safety factor for a) TRESCA b) VON MISES failure criteria



$\sigma_x = 30$ MPa $\quad \sigma_y = 150$ MPa $\quad \tau_{xy} = 80$ MPa

$$\sigma_{max,min} = \frac{30+150}{2} \pm \sqrt{\left(\frac{30-150}{2}\right)^2 + 80^2}$$

$\sigma_{max} = 180$ MPa $\quad \sigma_{min} = -10$ MPa

$\sigma_1 = 180 \quad \sigma_2 = 0 \quad \sigma_3 = -10$ MPa

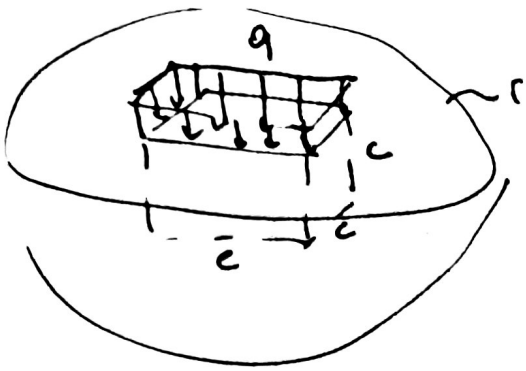
$$\sigma_n = \frac{\sigma_m}{n} \qquad n > 1$$

a) $\sigma_h = \sigma_1 - \sigma_3 \qquad \sigma_h = 180 - (-10) = 200$

$$n = \frac{350}{200} = 1{,}75$$

b) $\sigma_h^2 = 180^2 + (-10)^2 - (-180.10) = 38100 \qquad \sigma_h = 195.20$ MPa

$$n = \frac{350}{195{,}2} = 1{,}79$$



An elastic cube is embedded into a rigid body. The yield strength in ten of the material is $\tau_m = 12$ MPa and in compression $\tau_m' = 20$
Find q using Coulom yield criterion and the change in the length of the cube
$\nu = 0.25$ $c = 100$ mm $E = 80$ GPa

$\tau_z = -q$ $\varepsilon_x = \varepsilon_y = 0$

$$\varepsilon_x = \frac{1}{E}\left(\tau_x - \nu(\tau_y + \tau_z)\right) = \frac{1}{E}\left(\tau_x - 0{,}25(\tau_y - q)\right) = 0 \quad (1)$$

$$\varepsilon_y = \frac{1}{E}\left(\tau_y - \nu(\tau_x + \tau_z)\right) = \frac{1}{E}\left(\tau_y - 0{,}25(\tau_x - q)\right) = 0 \quad (2)$$

Solving (1) and (2)

$$\tau_x = \tau_y = -\frac{1}{3}q$$

Coulomb yield criterion

$$(\tau_{max} - \tau_{min}) = 2a - b(\tau_{max} + \tau_{min})$$

$$a = \frac{\tau_m' \tau_m}{\tau_m' + \tau_m} = \frac{20 \cdot 12}{20 + 12} = 7.5 \qquad b = \frac{\tau_m' - \tau_m}{\tau_m' + \tau_m} = \frac{20 - 12}{32} = 0.25$$

$$\tau_{max} = -\frac{1}{3}q \qquad \tau_{min} = -q$$

$$\left(-\frac{1}{3}q\right) - (-q) = 15 - 0{,}25\left[-\frac{q}{3} + \left(-\frac{q}{3}\right)\right] \qquad q = 45 \text{ MPa}$$

$\tau_x = \tau_y = -15$ MPa $\tau_z = -45$ MPa

$$\varepsilon_z = \frac{1}{80000}\left(-45 - 0{,}25(-15 - 15)\right) = -4{,}69\,10^{-4}$$

$$\Delta c = c \cdot \varepsilon_z = 0{,}0469 \text{ mm}$$